A single kinetic defect in a run-and-tumble ring: exact spectrum, exceptional points, and relaxation crossover

Abstract. We solve the spectral problem for one continuous-time run-and-tumble particle on an n-site ring when the tumbling rate at one site is ω_0 and the bulk rate is ω. The defect changes the $2n \times 2n$ generator by rank one. The characteristic polynomial is the homogeneous polynomial plus $(\omega_0 - \omega)/n$ times its ω-derivative. This identity gives Chebyshev and transfer-matrix quantization formulas, a parity factorization, and finite polynomial tests for every Jordan block. We separately resolve the singular flat-band point $\omega = 1$, where the eigenvalue -2 can carry Jordan blocks of sizes two or three. The infinite-line Green function gives an exact criterion for exponentially localized defect modes. For fixed positive bulk rate, reflection protects one homogeneous long-wave mode; consequently a slow defect, including $\omega_0 \asymp n^{-1}$, does not produce a new spectral-gap scaling. The nontrivial n^{-1} crossover occurs when the bulk rate itself is of order n^{-1}. Finally, the standard telegraph equation misses the lattice Poisson-diffusion term at fixed ω, while a diffusion-corrected telegraph equation agrees through order n^{-2}.

Keywords. Run-and-tumble particle; kinetic defect; non-Hermitian spectrum; exceptional point; localized mode; spectral gap; telegraph equation.

1. Introduction

Persistent random walks provide a microscopic description of motion with a finite directional memory. Continuum formulations with space-dependent turning statistics arise, for example, in the theory of chemotactic random walks [Sch93], while lattice run-and-tumble models retain the individual hopping and reversal events and connect directly to nonequilibrium particle systems [TTCB11]. For a homogeneous ring, the one-particle spectrum, its exceptional points, and the associated relaxation transition are known explicitly [MBE19].

This paper asks how that spectral picture changes when the tumbling rate is altered at one site. The inhomogeneity is local but the generator is non-normal, so eigenvalue crossings, Jordan structure, and long-wave relaxation must be tracked together. The central observation is that the defect is rank one in the polarization channel. It leads to a finite characteristic identity, a parity decomposition, an infinite-line Green-function criterion including a compact defect mode, and a direct comparison between lattice and telegraph relaxation. The fixed-rate and bulk-critical limits separate the effects of a local defect from those of a vanishing bulk tumbling rate.

2. Model and notation

Definition 2.1 (Defective run-and-tumble generator)**.** Let $n \geq 2$, let $\omega > 0$, and let $\omega_0 \geq 0$. Sites are $j \in \mathbb{Z}/n\mathbb{Z}$, orientations are $\sigma \in \{+, -\}$, and

$$\omega_j = \begin{cases} \omega_0, & j = 0, \\ \omega, & j \neq 0. \end{cases}$$

The forward generator $L = L_n(\omega, \omega_0)$ is

$$(Lu)_j^+ = u_{j-1}^+ - (1+\omega_j)u_j^+ + \omega_j u_j^-, \tag{2.1}$$

$$(Lu)_j^- = u_{j+1}^- - (1+\omega_j)u_j^- + \omega_j u_j^+. \tag{2.2}$$

Thus hopping in the current orientation has rate one and tumbling has rate ω_j. Put

$$\delta = \omega_0 - \omega, \qquad q_k = \frac{2\pi k}{n}, \qquad c_k = \cos q_k, \qquad s_k = \sin q_k.$$

Equations (2.1)–(2.2) use the same one-particle convention as equations (6)–(10) of [MBE19]. That paper records the homogeneous eigenvalues in its equation (10), the homogeneous exceptional points in equation (13), and the first-mode relaxation rate in equations (20)–(21).

Lemma 2.2 *(Rank-one defect)***.** *Let $L^{(0)} = L_n(\omega, \omega)$. If $d = e_{0,+} - e_{0,-}$, then*

$$L = L^{(0)} - \delta\, dd^{\mathsf{T}}.$$

In particular, the perturbation has rank one when $\delta \neq 0$.

Proof. At site zero, changing ω to ω_0 adds

$$\delta \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} = -\delta \begin{pmatrix} 1 \\ -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \end{pmatrix}.$$

Every other matrix entry is unchanged. ∎

3. Three exact characteristic equations

Definition 3.1 (Homogeneous factors)**.** For $\lambda \in \mathbb{C}$, set

$$a = \lambda + 1 + \omega, \tag{3.1}$$

$$x = \frac{a^2 + 1 - \omega^2}{2a}, \tag{3.2}$$

$$D_k(\lambda) = (\lambda + 1 - c_k)(\lambda + 1 + 2\omega - c_k) + s_k^2 = a^2 + 1 - \omega^2 - 2ac_k. \tag{3.3}$$

Although x is undefined at $a = 0$, each expression below containing $a^n T_n(x)$ is understood by its unique polynomial continuation.

Lemma 3.2 *(Homogeneous polynomial)***.** *The homogeneous characteristic polynomial is*

$$\begin{aligned} P_n^{(0)}(\lambda;\omega) &= \det(\lambda I - L^{(0)}) \\ &= \prod_{k=0}^{n-1} D_k(\lambda) \qquad (3{\cdot}4) \\ &= 2a^n\big(T_n(x) - 1\big), \qquad (3{\cdot}5) \end{aligned}$$

where T_n is the Chebyshev polynomial of the first kind. The two roots of D_k are

$$\lambda_k^{\pm} = c_k - 1 - \omega \pm \sqrt{\omega^2 - s_k^2}.$$

Proof. Fourier transformation gives the block

$$W_k = \begin{pmatrix} e^{\mathrm{i}q_k} - 1 - \omega & \omega \\ \omega & e^{-\mathrm{i}q_k} - 1 - \omega \end{pmatrix}.$$

Its characteristic polynomial is D_k, which proves (3·4) and the displayed roots. Since $D_k = 2a(x - c_k)$ and

$$\prod_{k=0}^{n-1}(x - c_k) = 2^{1-n}\big(T_n(x) - 1\big),$$

equation (3·5) follows. ∎

Theorem 3.3 *(Exact finite characteristic polynomial)***.** *For every $n \geq 2$, $\omega > 0$, and $\omega_0 \geq 0$,*

$$\begin{aligned} P_n(\lambda;\omega,\omega_0) &:= \det(\lambda I - L_n(\omega,\omega_0)) \qquad (3{\cdot}6) \\ &= P_n^{(0)}(\lambda;\omega) + \frac{\delta}{n}\,\partial_\omega P_n^{(0)}(\lambda;\omega) \qquad (3{\cdot}7) \\ &= P_n^{(0)}(\lambda;\omega)\left[1 + \frac{2\delta}{n}\sum_{k=0}^{n-1}\frac{\lambda + 1 - c_k}{D_k(\lambda)}\right]. \qquad (3{\cdot}8) \end{aligned}$$

Equation (3·8) is interpreted by polynomial continuation at a homogeneous eigenvalue. Equivalently,

$$P_n = 2a^n(T_n(x) - 1) + 2\delta a^{n-1}(T_n(x) - 1) + \delta a^{n-2}\lambda(\lambda + 2)U_{n-1}(x), \qquad (3{\cdot}9)$$

where U_{n-1} is the Chebyshev polynomial of the second kind and the right-hand side again denotes its polynomial continuation.

Proof. Let $B = \lambda I - L^{(0)}$. Lemma 2.2 and the matrix determinant lemma give

$$\det(\lambda I - L) = \det B\big(1 + \delta d^{\mathsf{T}} B^{-1} d\big).$$

In Fourier space,

$$d^{\mathsf{T}} B^{-1} d = \frac{1}{n}\sum_{k=0}^{n-1}\frac{2(\lambda + 1 - c_k)}{D_k(\lambda)},$$

which proves (3.8). Direct differentiation of (3.4) gives

$$\partial_\omega P_n^{(0)} \;=\; P_n^{(0)} \sum_k \frac{2(\lambda+1-c_k)}{D_k},$$

and hence (3.7). Finally,

$$\partial_\omega x \;=\; \frac{\lambda(\lambda+2)}{2a^2}, \qquad T_n'(x) \;=\; nU_{n-1}(x),$$

and differentiating (3.5) proves (3.9). ■

Theorem 3.4 *(Transfer-matrix quantization)***.** *For a local rate $r \geq 0$, define*

$$T_r(\lambda) \;=\; \frac{1}{\lambda+1+r}\begin{pmatrix} 1 & r \\ -r & (\lambda+1+r)^2 - r^2 \end{pmatrix}.$$

Then $\det T_r \;=\; 1$, and, away from the displayed denominators,

$$P_n(\lambda;\omega,\omega_0) \;=\; a^{n-1}(\lambda+1+\omega_0)\left\{\operatorname{tr}\left[T_\omega(\lambda)^{n-1}T_{\omega_0}(\lambda)\right] - 2\right\}.$$

After multiplication by the denominators, this identity holds for every λ by polynomial continuation. Thus an eigenvalue is equivalently characterized by

$$\operatorname{tr}\left[T_\omega(\lambda)^{n-1}T_{\omega_0}(\lambda)\right] \;=\; 2.$$

Moreover, if $t \;=\; \operatorname{tr} T_\omega \;=\; 2x$, then

$$T_\omega^{n-1} \;=\; U_{n-2}(x)T_\omega - U_{n-3}(x)I,$$

with $U_{-1} \;=\; 0$.

Proof. The eigenvalue equations at site j are

$$\begin{aligned}(\lambda+1+\omega_j)u_j^+ - \omega_j u_j^- \;&=\; u_{j-1}^+,\\ (\lambda+1+\omega_j)u_j^- - \omega_j u_j^+ \;&=\; u_{j+1}^-.\end{aligned}$$

Consequently

$$\begin{pmatrix} u_j^+ \\ u_{j+1}^- \end{pmatrix} \;=\; T_{\omega_j}(\lambda)\begin{pmatrix} u_{j-1}^+ \\ u_j^- \end{pmatrix}.$$

Periodic boundary conditions require the product monodromy to have eigenvalue one. Its determinant is one, so this is equivalent to trace two. To identify the polynomial, retain temporarily arbitrary rates $r_0, \ldots, r_{n-1}$, denote the resulting generator by $L_n(r_0, \ldots, r_{n-1})$, and put $b_j \;=\; \lambda+1+r_j$. Gaussian elimination of the $2n$ eigenvalue equations, in the order $u_0^+, u_1^+, \ldots, u_{n-1}^+$, has pivots $b_0, \ldots, b_{n-1}$. If $M \;=\; T_{r_{n-1}} \cdots T_{r_0}$, the two remaining cyclic boundary equations have coefficient matrix $M - I$. The cyclic row ordering contributes one minus sign, and $-\det(M-I) \;=\; \operatorname{tr} M - 2$ because $\det M \;=\; 1$. Consequently

$$\det(\lambda I - L_n(r_0, \ldots, r_{n-1})) \;=\; \left(\prod_{j=0}^{n-1} b_j\right)\left[\operatorname{tr}(T_{r_{n-1}} \cdots T_{r_0}) - 2\right]. \tag{3.10}$$

This is an identity of rational functions where all $b_j \neq 0$; multiplication by the displayed pivots makes both sides polynomials, so the identity extends to every λ. Taking $r_0 = \omega_0$ and all other rates equal to ω, and using cyclicity of the trace, gives (3.4), including algebraic multiplicities. Equation (3.4) is the Cayley–Hamilton identity for a unimodular 2×2 matrix. ■

4. Parity factorization and every finite exceptional point

Definition 4.1 (Kinetic reflection)**.** Define the involution

$$(Ru)_j^+ = u_{-j}^-, \qquad (Ru)_j^- = u_{-j}^+.$$

Write $\mathcal{H}_\pm = \ker(R \mp I)$.

Lemma 4.2 *(Protected sector)*. *The generator commutes with R, the defect vector satisfies $Rd = -d$, and*

$$L|_{\mathcal{H}_+} = L^{(0)}|_{\mathcal{H}_+}.$$

Thus the defect changes only the reflection-odd sector.

Proof. Reflection reverses position and orientation simultaneously, leaving the directed hopping rules and the rate field ω_j invariant. At the fixed site zero it exchanges the two entries of d, hence $Rd = -d$. Lemma 2.2 then vanishes on $\mathcal{H}_+$. ■

Definition 4.3 (Parity polynomials)**.** Put $m = \lfloor (n-1)/2 \rfloor$ and let ε_n be one for even n and zero for odd n. Define

$$H_{n,+}(\lambda) = \lambda(\lambda+2)^{\varepsilon_n} \prod_{k=1}^{m} D_k(\lambda), \tag{4.1}$$

$$H_{n,-}^{(0)}(\lambda) = (\lambda+2\omega)(\lambda+2+2\omega)^{\varepsilon_n} \prod_{k=1}^{m} D_k(\lambda), \tag{4.2}$$

$$G_n(\lambda) = \frac{1}{n}\left[\frac{1}{\lambda+2\omega} + \frac{\varepsilon_n}{\lambda+2+2\omega} + 2\sum_{k=1}^{m} \frac{\lambda+1-c_k}{D_k(\lambda)}\right], \tag{4.3}$$

$$Q_{n,-}(\lambda) = H_{n,-}^{(0)}(\lambda)\big(1 + 2\delta G_n(\lambda)\big). \tag{4.4}$$

Every quotient in (4.4) cancels, so $Q_{n,-}$ is a monic polynomial of degree n. An entirely polynomial version is

$$Q_{n,-} = H_{n,-}^{(0)} + \frac{2\delta}{n}\left[\frac{H_{n,-}^{(0)}}{\lambda+2\omega} + \varepsilon_n \frac{H_{n,-}^{(0)}}{\lambda+2+2\omega} + 2\sum_{k=1}^{m}(\lambda+1-c_k)\frac{H_{n,-}^{(0)}}{D_k}\right]. \tag{4.5}$$

Theorem 4.4 *(Exact parity factorization).* *The characteristic polynomial factors as*

$$P_n(\lambda;\omega,\omega_0) \;=\; H_{n,+}(\lambda)Q_{n,-}(\lambda).$$

The first factor is the characteristic polynomial on $\mathcal{H}_+$; the second is the characteristic polynomial on $\mathcal{H}_-$.

Proof. The modes q_k and $-q_k$ form a four-dimensional Fourier subspace, which splits into two isomorphic two-dimensional R-eigenspaces. Each has characteristic polynomial D_k. At $q = 0$, the vectors $(1,1)$ and $(1,-1)$ have parities $+$ and $-$ and eigenvalues 0 and -2ω. When n is even, the $q = \pi$ density and polarization vectors have parities $+$ and $-$ and eigenvalues -2 and $-2-2\omega$. This gives (4.1)–(4.2). Lemma 4.2 and the rank-one determinant lemma restricted to $\mathcal{H}_-$ give (4.4). Multiplication proves (4.4). ∎

Definition 4.5 (Exceptional point)**.** For an eigenvalue μ, let $a(\mu)$ be its algebraic multiplicity and $g(\mu) = \dim\ker(\mu I - L)$ its geometric multiplicity. We call (ω,ω_0,μ) an exceptional point when $a(\mu) > g(\mu)$. Write $J_r(\mu)$ for a Jordan block of size r at μ.

Lemma 4.6 *(No inter-wave coincidences away from the flat band).* *If $c_k \neq c_\ell$ and $D_k(\mu) = D_\ell(\mu) = 0$, then $\omega = 1$ and $\mu = -2$.*

Proof. Subtracting the two equations gives $2a(c_k - c_\ell) = 0$, hence $a = 0$. Substitution into (3.3) gives $1-\omega^2 = 0$. Since $\omega > 0$, this means $\omega = 1$ and then $\mu = -1-\omega = -2$. ∎

Theorem 4.7 *(Jordan classification for $\omega \neq 1$).* *Assume $\omega \neq 1$.*

1. *In the protected sector, the endpoint factors in (4.1) give one-dimensional blocks. For each $1 \le k \le m$, a root of D_k gives one J_1, except that*

$$\omega = |s_k|, \qquad \mu = c_k - 1 - \omega. \tag{4.6}$$

   gives one $J_2(\mu)$.

2. *In the affected sector, if*

$$Q_{n,-}(\mu) = Q'_{n,-}(\mu) = \cdots = Q^{(r-1)}_{n,-}(\mu) = 0, \qquad Q^{(r)}_{n,-}(\mu) \neq 0, \tag{4.7}$$

   then the complete affected-sector contribution at μ is one block $J_r(\mu)$.

3. *The full Jordan form at μ is the direct sum of the blocks in items 1 and 2. In particular, a crossing between the two parity sectors is semisimple unless one of its sector blocks has size greater than one.*

Consequently, (4.6) together with (4.7) is an exact finite criterion for every exceptional point when $\omega \neq 1$. The order r in (4.7) classifies exceptional points of arbitrary order.

Proof. The discriminant of D_k is $4(\omega^2 - s_k^2)$. At a zero discriminant its Fourier block is not scalar and hence is a single J_2; otherwise each root is simple. Lemma 4.6 excludes coincidences between distinct c_k values.

It remains to show that the affected restriction is cyclic. In each two-dimensional homogeneous parity block, the spectral weight of the defect vector has numerator $2(\lambda+1-c_k)$. This numerator and D_k have no common root for an interior wave number: a common root would give $\lambda+1 = c_k$, after which $D_k = s_k^2 \neq 0$. The endpoint weights in (4.3) are also nonzero. Lemma 4.6 makes the block minimal polynomials coprime. The defect vector is therefore cyclic for the homogeneous odd restriction. If A has cyclic vector d, then d is also cyclic for $A - \delta dd^{\mathsf{T}}$, because each $(A-\delta dd^{\mathsf{T}})^j d$ equals $A^j d$ plus a linear combination of lower powers. Hence the minimal and characteristic polynomials of the affected restriction both equal $Q_{n,-}$. A cyclic matrix has one Jordan block for each eigenvalue, of size equal to the root order. The direct-sum assertion follows from the invariant parity splitting. ∎

Theorem 4.8 *(The complete flat-band classification at $\omega = 1$)***.** *Let $\omega = 1$ and put $\mu = -2$.*

1. *If $\omega_0 = 1$ and $4 \nmid n$, the contribution at μ is $nJ_1(\mu)$.*
2. *If $\omega_0 = 1$ and $4 \mid n$, the contribution is*

$$2J_2(\mu) \oplus (n-2)J_1(\mu).$$

3. *If $\omega_0 \neq 1$ and $n \equiv 1$ or $2 \pmod 4$, the contribution is $(n-1)J_1(\mu)$.*
4. *If $\omega_0 \neq 1$ and $n \equiv 0 \pmod 4$, the contribution is*

$$J_2(\mu) \oplus (n-2)J_1(\mu).$$

5. *If $n \equiv 3 \pmod 4$, $\omega_0 \neq 1$, and $\omega_0 \neq (n-1)/(n+1)$, the contribution is again $J_2(\mu) \oplus (n-2)J_1(\mu)$.*
6. *If $n \equiv 3 \pmod 4$ and $\omega_0 = (n-1)/(n+1)$, the contribution is*

$$J_3(\mu) \oplus (n-2)J_1(\mu).$$

For eigenvalues $\lambda \neq -2$, every protected-sector root is a one-dimensional block. Remove the exact power of $(\lambda+2)$ specified above from $Q_{n,-}$. The remaining affected polynomial is cyclic, so the derivative test (4.7) applied to that reduced polynomial classifies every remaining affected block at $\omega = 1$. The full Jordan form is the direct sum of the protected and affected blocks.

Proof. Write $t = \lambda + 2$. At $\omega = 1$,

$$D_k = t(t-2c_k), \qquad P_n = t^{n-1}F_n(t),$$

where, for $\delta = \omega_0 - 1$,

$$F_n(t) = 2t\big(T_n(t/2)-1\big) + \delta\left[2\big(T_n(t/2)-1\big) + (t-2)U_{n-1}(t/2)\right]. \tag{4.8}$$

The values

$$T_n(0) = \cos(n\pi/2), \qquad U_{n-1}(0) = \sin(n\pi/2)$$

give the following algebraic multiplicities. For $\delta = 0$ they are n, except that $4 \mid n$ gives $n+2$. For $\delta \neq 0$ they are $n-1$ when $n \equiv 1, 2 \pmod 4$, and n when $n \equiv 0, 3 \pmod 4$. In the last congruence class, $F_n'(0) = -2 - \delta(n+1)$; it vanishes exactly at $\delta = -2/(n+1)$, when $F_n''(0) \neq 0$ and the multiplicity is $n+1$.

The bulk equations at $t = 0$ read

$$u_j^- = -u_{j-1}^+ \qquad (j \neq 0).$$

Together they impose this relation for every j. The defect equation adds no constraint when $\omega_0 = 1$ and otherwise adds the single constraint $u_0^+ + u_{n-1}^+ = 0$. Thus the geometric multiplicity is n in the homogeneous case and $n-1$ otherwise. Fourier decomposition shows that the two excess algebraic dimensions when $4 \mid n$ and $\delta = 0$ are the two $q = \pm\pi/2$ blocks, each a J_2. For $\delta \neq 0$, the parity factorization places the $4 \mid n$ excess in one protected J_2.

It remains to distinguish one J_3 from two J_2 blocks in the tuned $n \equiv 3 \pmod 4$ case. Put $N_0 = L_n(1,1) + 2I$ and $N = L_n(1,\omega_0) + 2I = N_0 - \delta d d^{\mathsf{T}}$. Because $4 \nmid n$, zero is a semisimple eigenvalue of N_0 with eigenspace $E = \ker N_0$ of dimension n. Let Π be its spectral projection. Write

$$\det(tI - N_0) = t^n h(t), \qquad h(0) \neq 0.$$

Semisimplicity gives $(tI - N_0)^{-1} = t^{-1}\Pi + O(1)$. The rank-one determinant formula then shows that the coefficient of t^{n-1} in $\det(tI - N)$ is $\delta h(0) d^{\mathsf{T}}\Pi d$. The multiplicity computation above shows that this coefficient vanishes for $n \equiv 3 \pmod 4$, and therefore

$$d^{\mathsf{T}}\Pi d = 0. \tag{4.9}$$

The eigenspace of N is $K = E \cap \ker d^{\mathsf{T}}$, which has dimension $n-1$. If $v \in K \cap \operatorname{ran} N$, write $v = Nu$ and apply Π:

$$v = \Pi N u = -\delta\, \Pi d\, (d^{\mathsf{T}} u). \tag{4.10}$$

Thus $K \cap \operatorname{ran} N$ has dimension at most one. It has dimension one because the algebraic multiplicity exceeds the geometric multiplicity. Hence exactly one Jordan block at zero for N is nontrivial. It is a J_2 when the algebraic multiplicity is n, and the additional algebraic order at $\delta = -2/(n+1)$ makes that same block a J_3. This proves all six cases and excludes the alternative of two J_2 blocks in the tuned case.

After the flat factor is removed, the coprime-block cyclicity argument from Theorem 4.7 applies unchanged. ∎

Corollary 4.9 *(Finite algebraic exceptional set).* *For fixed n, all non-flat affected exceptional parameters lie on the explicit algebraic set*

$$\operatorname*{Res}_{\lambda}\big(Q_{n,-}, \partial_\lambda Q_{n,-}\big) = 0,$$

with $Q_{n,-}$ given by (4.5). At $\omega = 1$, the flat factor is removed before taking the resultant. Root orders and theorems 4.7–4.8 distinguish exceptional points from semisimple parity crossings without a numerical tolerance.

Proof. The resultant vanishes exactly when the two displayed polynomials share a root. The preceding two theorems identify the Jordan block attached to every such root and separately account for the flat factor. ∎

5. Defect-localized modes

Definition 5.1 (Bulk spectral set)**.** For the infinite translation-invariant chain, define

$$\Sigma_\omega = \left\{c - 1 - \omega \pm \sqrt{\omega^2 - (1 - c^2)} \;:\; c \in [-1, 1]\right\}.$$

For $\lambda \notin \Sigma_\omega$ with $a \neq 0$, retain a and x from (3.1)–(3.2), and choose

$$s(\lambda) = \sqrt{x(\lambda)^2 - 1}$$

as the branch satisfying $s(\lambda) \sim x(\lambda)$ as $|\lambda| \to \infty$. Put $z = x - s$, so $|z| < 1$.

Theorem 5.2 *(Infinite-line localization criterion)***.** *Let $\delta \neq 0$ and $\lambda \notin \Sigma_\omega$. First suppose $a \neq 0$. Then λ is an exponentially localized eigenvalue of the infinite chain with the same single tumbling defect if and only if*

$$1 + 2\delta g_\infty(\lambda) = 0, \tag{5.1}$$

where

$$g_\infty(\lambda) = \frac{1}{2a}\left(1 + \frac{\lambda + 1 - x}{s}\right) = \frac{z(\lambda + 1 - z)}{a(1 - z^2)}. \tag{5.2}$$

The localization length is

$$\xi(\lambda) = \frac{-1}{\log|z|}.$$

Equivalently, with $a = \lambda + 1 + \omega$, the pair (a, z) satisfies

$$a^2 + 1 - \omega^2 - a(z + z^{-1}) = 0, \tag{5.3}$$

$$a(1 - z^2) + 2\delta z(a - \omega - z) = 0, \qquad 0 < |z| < 1. \tag{5.4}$$

After elimination of a, every admissible z is a root of

$$\begin{aligned} 0 = {} & 2\delta z^3 + (2\delta\omega + 1 - \omega^2)z^2 \\ & + (4\delta^2\omega + 4\delta\omega^2 - 2\delta)z + (2\delta\omega + \omega^2 - 1), \end{aligned} \tag{5.5}$$

and, when the denominator is nonzero,

$$a = \frac{2\delta z(\omega + z)}{1 - z^2 + 2\delta z}, \qquad \lambda = a - 1 - \omega. \tag{5.6}$$

A root of (5.5) *is accepted only when* (5.1)*, or equivalently both* (5.3) *and* (5.4)*, is satisfied; this removes roots introduced by elimination or by clearing denominators.*

There is one additional compact case, at the point where $a = 0$*. If* $\omega \neq 1$*, then*

$$\lambda = -1-\omega \quad \text{is localized if and only if} \quad \omega_0 = \frac{\omega^2+1}{2\omega}. \tag{5.7}$$

Its decay root is $z = 0$*, its localization length is* $\xi = 0$*, and its eigenvector is supported at the defect and its two neighboring sites. Thus the admissible range in the preceding formulas is* $0 < |z| < 1$*;* $z = 0$ *is accepted precisely in* (5.7)*.*

Proof. The polarization entry of the bulk resolvent at the defect site is

$$\begin{aligned} g_\infty(\lambda) &= \frac{1}{2\pi}\int_0^{2\pi} \frac{\lambda+1-\cos q}{a^2+1-\omega^2-2a\cos q}\,dq \\ &= \frac{1}{2a}\left[1+(\lambda+1-x)\frac{1}{2\pi}\int_0^{2\pi}\frac{dq}{x-\cos q}\right] \\ &= \frac{1}{2a}\left(1+\frac{\lambda+1-x}{s}\right). \end{aligned}$$

The rank-one eigenvalue equation is (5.1). Since $x = (z+z^{-1})/2$ and $s = (z^{-1}-z)/2$, elementary simplification gives the second expression in (5.2), and the Green kernel decays as $z^{|j|}$. Clearing the denominator in (5.1) gives (5.4); the bulk dispersion relation gives (5.3). Eliminating a gives (5.5), while solving (5.4) for a gives (5.6).

It remains to evaluate the point omitted by division by a. At $\lambda = -1-\omega$ the Fourier denominator is the nonzero constant $1-\omega^2$, and hence

$$g_\infty(-1-\omega) = \frac{1}{2\pi}\int_0^{2\pi}\frac{-\omega-\cos q}{1-\omega^2}\,dq = \frac{\omega}{\omega^2-1}.$$

The secular equation is therefore equivalent to $\delta = (1-\omega^2)/(2\omega)$, which is exactly (5.7). In Fourier variables the associated resolvent vector is proportional to

$$\begin{pmatrix} -\omega - e^{-\mathrm{i}q} \\ \omega + e^{\mathrm{i}q} \end{pmatrix}.$$

Its inverse Fourier transform is supported at the defect and its two neighbors. This proves the compact case and also shows directly that it is the limiting root $z = 0$. ∎

Proposition 5.3 *(Finite-ring convergence of a localized mode)***.** *Suppose* (5.1) *has a simple root* λ_∞ *with decay root* z_∞*,* $0 < |z_\infty| < 1$*. Then, for all sufficiently large* n*, the finite polynomial* $Q_{n,-}$ *has a unique root* λ_n *near* λ_∞*, and*

$$\lambda_n - \lambda_\infty = O(|z_\infty|^n).$$

The corresponding right eigenvector has site mass bounded by $C|z_\infty|^{2\,\mathrm{dist}(j,0)}$ *after normalization. In the compact case* (5.7)*, the infinite and finite eigenvalues agree exactly and the eigenvector is supported where* $\mathrm{dist}(j,0) \leq 1$*.*

Proof. The root-of-unity sum of the bulk Green kernel is the exact identity

$$G_n(\lambda) \;=\; \frac{1}{2a}\left[1+\frac{\lambda+1-x}{s}\frac{1+z^n}{1-z^n}\right]. \tag{5.8}$$

Indeed, Fourier expansion of $(x-\cos q)^{-1}$ gives

$$\frac{1}{n}\sum_{k=0}^{n-1}\frac{1}{x-\cos q_k} \;=\; \frac{1}{s}\frac{1+z^n}{1-z^n}.$$

Consequently

$$G_n - g_\infty \;=\; \frac{\lambda+1-x}{as}\,\frac{z^n}{1-z^n}. \tag{5.9}$$

Choose a closed disk B about λ_∞ disjoint from Σ_ω on which $|z(\lambda)| \le r < 1$ and containing no other zero of $1+2\delta g_\infty$. Equation (5.9) and its λ-derivative are bounded on B by $C_B r^n$. Rouché's theorem on ∂B, followed by the simple-zero estimate at λ_∞, gives one zero λ_n. Evaluation of (5.9) at λ_∞, together with $z(\lambda_n) = z_\infty + O(\lambda_n - \lambda_\infty)$, gives $\lambda_n - \lambda_\infty = O(|z_\infty|^n)$.

The Fourier coefficients of the two bulk resolvent entries are linear combinations of $z^{|j|}$ and $z^{|j\pm1|}$. Their periodizations are bounded by

$$\frac{r^{\operatorname{dist}(j,0)} + r^{n-\operatorname{dist}(j,0)}}{1-r^n}.$$

The coefficients converge to their nonzero infinite-volume values, so normalization changes this bound by a fixed factor. Using $z(\lambda_n) = z_\infty + O(|z_\infty|^n)$ gives the stated site-mass estimate.

In the compact case, the Laurent-polynomial Fourier vector displayed in the proof of Theorem 5.2 is an exact finite-ring eigenvector for every n (with the two neighbors identified when $n = 2$), proving the last assertion. ∎

6. Large-ring spectral gap

Definition 6.1 (Spectral gap)**.** For $\omega > 0$ and $n \ge 2$, irreducibility makes zero a simple eigenvalue. Define

$$\gamma_n(\omega,\omega_0) \;=\; -\max\{\Re\lambda \,:\, \lambda \in \operatorname{spec} L_n(\omega,\omega_0),\ \lambda \neq 0\}.$$

Put $q = 2\pi/n$ and

$$\gamma_n^{\mathrm{hom}}(\omega) \;=\; 1-\cos q+\omega-\sqrt{\omega^2-\sin^2 q}, \tag{6.1}$$

where the square root is positive once n is sufficiently large for fixed $\omega > 0$.

Lemma 6.2 *(Protected first mode and affected pole)***.** *For every n, the protected sector contains $\lambda_1^+ = -\gamma_n^{\mathrm{hom}}$. For fixed positive ω and ω_0, the affected root born from the same homogeneous pole is*

$$\lambda_{1,-} \;=\; \lambda_1^+ + \frac{\delta(1+\omega)}{\omega^2(1+\omega_0)}\frac{q^2}{n} + O(n^{-4}). \tag{6.2}$$

Proof. The first assertion is Theorem 4.4. Near λ_1^+, the two terms $k = 1, n-1$ in the full Fourier Green function have residue

$$R_1 = \frac{\lambda_1^+ + 1 - c_1}{\lambda_1^+ - \lambda_1^-} = -\frac{\omega - r_1}{2r_1}, \qquad r_1 = \sqrt{\omega^2 - \sin^2 q}.$$

Thus

$$G_n(\lambda) = \frac{2R_1}{n(\lambda - \lambda_1^+)} + G_n^{\mathrm{reg}}(\lambda).$$

At $\lambda = 0$, every nonzero-wave summand is exactly $1/[2(1+\omega)]$. Splitting the sum into $|q_k| \le n^{-1/2}$ and its complement and using $\lambda_1^+ = O(n^{-2})$ gives

$$G_n^{\mathrm{reg}}(\lambda_1^+) = \frac{1}{2(1+\omega)} + O(n^{-1}), \qquad (G_n^{\mathrm{reg}})'(\lambda) = O(n)$$

for $|\lambda - \lambda_1^+| = O(n^{-3})$. Solving $1 + 2\delta G_n = 0$ gives

$$\lambda_{1,-} - \lambda_1^+ = \frac{2\delta(\omega - r_1)}{nr_1} \frac{1}{1 + \delta/(1+\omega)} + O(n^{-4}),$$

which expands to (6.2). ∎

Lemma 6.3 *(Right-edge root separation).* *Fix $\omega > 0$ and $\omega_0 \ge 0$, and put $D = (1+\omega)/(2\omega)$. There is N such that, for $n \ge N$, every nonzero eigenvalue other than the protected and affected roots born from $q = \pm 2\pi/n$ satisfies*

$$\Re\lambda \le \Re\lambda_1^+ - \pi^2 D n^{-2}. \tag{6.3}$$

The constants can be chosen uniformly when (ω, ω_0) ranges over a compact set with ω bounded away from zero.

Proof. We give the root count because a local pole expansion alone does not identify the spectral gap. The slow homogeneous dispersion has the uniform Taylor estimate

$$\lambda^+(q) = -Dq^2 + O(q^4). \tag{6.4}$$

Choose $q_* > 0$ so that the remainder in (6.4) has modulus at most $Dq^2/4$ for $|q| \le q_*$. On the compact part $q_* \le |q| \le \pi$, continuity of the two bulk branches gives

$$\eta_* = -\frac{1}{2} \max_{q_* \le |q| \le \pi,\, \pm} \Re\lambda^\pm(q) > 0.$$

If $\delta = 0$, the affected roots are the homogeneous roots and the claim follows directly from (6.4). Suppose $\delta \ne 0$. At the right bulk edge,

$$\lim_{\lambda \to 0,\, \lambda \notin \Sigma_\omega} \big(1 + 2\delta g_\infty(\lambda)\big) = 1 + \frac{\delta}{1+\omega} = \frac{1+\omega_0}{1+\omega} > 0. \tag{6.5}$$

On a compact parameter set this margin has a positive minimum.

For the affected polynomial, use the Fourier partial-fraction formula (4.3). On a circle about a slow pole λ_k^+ of radius Aq_k^2/n, its polar term is $2R_k/[n(\lambda - \lambda_k^+)]$, where $R_k = O(q_k^2)$, while the sum with that pole removed is uniformly bounded after the small-wave terms are paired. The split used in Lemma 6.2 gives more precisely

$$G^{\mathrm{reg}}_{n,k}(\lambda) \;=\; \frac{1}{2(1+\omega)} + O(q_* + n^{-1})$$

throughout the pole cell bounded by the midpoints to the adjacent slow poles. Decrease q_* and increase N so that $|1 + 2\delta G^{\mathrm{reg}}_{n,k}|$ is at least half the minimum in (6.5).

Multiply the secular equation by $\lambda - \lambda_k^+$. On the boundary of the pole cell, whose distance from λ_k^+ is comparable to q_k/n, the constant polar contribution is smaller by $O(q_*)$ than the nonvanishing linear term. Rouché's theorem therefore counts exactly one affected zero in each cell. Applying it again on a circle of radius Aq_k^2/n, with fixed sufficiently large A, localizes that zero at

$$\lambda_{k,-} \;=\; \lambda_k^+ + O(q_k^2/n) \qquad (0 \;<\; |q_k| \;\le\; q_*). \tag{6.6}$$

The pole cells account for the extended zeros in $|\lambda| \;<\; \eta_*$. The exact Green formula (5.8) and the nonzero limit (6.5) show that localized zeros are separated from zero. Compactness supplies the same separation uniformly on the stated parameter sets.

For $k \;\ge\; 2$, equations (6.4) and (6.6) give, after increasing N,

$$\Re\lambda_k^+,\; \Re\lambda_{k,-} \;\le\; -3Dq_k^2/4 + Dq_k^2/8 \;\le\; -5Dq_1^2/2.$$

The first protected and affected roots are $-Dq_1^2 + O(n^{-3})$. The compact part of the spectrum lies to the left of $-\eta_*$. These bounds imply (6.3). Finally, the parity factorization has degree n in each sector, so the pole-circle count and the compact count account for all $2n$ roots. ∎

Theorem 6.4 *(Fixed-rate gap asymptotics)***.** *Fix $\omega \;>\; 0$ and $\omega_0 \;\ge\; 0$.*

1. *If $\omega_0 \;\le\; \omega$, then for every sufficiently large n,*

$$\gamma_n(\omega,\omega_0) \;=\; \gamma_n^{\mathrm{hom}}(\omega). \tag{6.7}$$

2. *If $\omega_0 \;>\; \omega$, then*

$$\gamma_n(\omega,\omega_0) \;=\; \gamma_n^{\mathrm{hom}}(\omega) - \frac{4\pi^2(\omega_0-\omega)(1+\omega)}{\omega^2(1+\omega_0)}\, n^{-3} + O(n^{-4}). \tag{6.8}$$

3. *In both cases,*

$$\begin{aligned}\gamma_n^{\mathrm{hom}}(\omega) \;=\;& \frac{2\pi^2(1+\omega)}{\omega}\, n^{-2} \\ & - 16\pi^4\left(\frac{1}{24} + \frac{1}{6\omega} - \frac{1}{8\omega^3}\right) n^{-4} + O(n^{-6}).\end{aligned} \tag{6.9}$$

The errors are uniform when (ω,ω_0) ranges over a compact set with ω bounded away from zero; in (6.7) the ordering is uniform away from $\omega_0 \;=\; \omega$.

Proof. Taylor expansion of (6.1) gives (6.9). Lemma 6.2 shows that the affected first root moves toward zero precisely when $\delta > 0$. Lemma 6.3 shows that no other extended, band-edge, or localized root can compete with this first pair. Therefore the protected root wins when $\delta \le 0$, proving (6.7), and the affected root wins when $\delta > 0$. Substitution of $q^2 = 4\pi^2 n^{-2}$ in (6.2) then gives (6.8). ∎

Corollary 6.5 *(Fixed ratio, vanishing ratio, and a slow-site scaling)***.** *Let the bulk rate $\omega > 0$ be fixed.*

1. *If $\omega_0/\omega \to \rho \in (0,\infty)$, equations (6.7)–(6.9) apply. For $\rho < 1$ the protected branch is selected; for $\rho > 1$ the affected branch and its n^{-3} term are selected. The boundary $\rho = 1$ is decided by the eventual sign and size of $\omega_0 - \omega$.*
2. *If $\omega_0/\omega \to 0$, then (6.7) holds for every sufficiently large n.*
3. *In particular, if $\omega_0 \sim \beta/n$ with $\beta \ge 0$, then (6.7) still holds. A single slow tumbling site therefore has no separate $\omega_0 \asymp n^{-1}$ gap crossover when the bulk rate is fixed.*

Proof. In items 2 and 3 one has $\omega_0 < \omega$ for every sufficiently large n, so Theorem 6.4 applies. Item 1 follows by the same sign test and compact-uniform expansion. ∎

Theorem 6.6 *(Bulk-critical double scaling)***.** *Set*

$$\omega = \frac{\alpha}{n}, \qquad \omega_0 = \frac{\beta}{n}, \qquad k = 2\pi,$$

where $\alpha > 0$ and $\beta \ge 0$ are fixed. Away from $\alpha = k$,

$$\alpha < k: \quad \gamma_n = \frac{\alpha}{n} + \left[\frac{k^2}{2} - 2(\alpha-\beta)_+\right] n^{-2} + O(n^{-3}), \tag{6.10}$$

$$\alpha > k: \quad \gamma_n = \frac{\alpha - \sqrt{\alpha^2 - k^2}}{n} + \left[\frac{k^2}{2} - 2(\beta-\alpha)_+ \frac{\alpha - \sqrt{\alpha^2-k^2}}{\sqrt{\alpha^2-k^2}}\right] n^{-2} + O(n^{-3}). \tag{6.11}$$

At $\alpha = k$,

$$\beta \le k: \quad \gamma_n = \frac{k}{n} + k^2\left(\frac{1}{2} - \frac{1}{\sqrt{3}}\right) n^{-2} + O(n^{-3}), \tag{6.12}$$

$$\beta > k: \quad \gamma_n = \frac{k}{n} - 2\sqrt{k(\beta-k)}\, n^{-3/2} + O(n^{-2}). \tag{6.13}$$

Thus the first homogeneous exceptional scaling is a square-root boundary layer for a faster defect, while the side selecting the affected parity mode reverses as α passes through 2π.

Proof. For $\alpha < k$, the protected first pair has

$$\Re\lambda_1^+ = -\frac{\alpha}{n} - \frac{k^2}{2n^2} + O(n^{-3}).$$

The pole residue tends to $R_1 = \frac{1}{2} + \mathrm{i}\alpha/(2\sqrt{k^2-\alpha^2})$, and $\delta = (\beta-\alpha)/n$. Hence the affected real part shifts by $-2(\beta-\alpha)n^{-2} + O(n^{-3})$. Taking the larger real part gives (6.10).

For $\alpha > k$, put $s = \sqrt{\alpha^2-k^2}$. The protected rate is $(\alpha-s)n^{-1} + k^2(2n^2)^{-1} + O(n^{-3})$, while $R_1 \to -(\alpha-s)/(2s)$. The same pole equation shifts the affected eigenvalue by $2(\beta-\alpha)(\alpha-s)/(sn^2) + O(n^{-3})$. Selection of the larger real part gives (6.11). The first wave number remains separated at order n^{-1} from all higher wave numbers in this case. For $\alpha < k$, the mode with fixed wave index $\ell \geq 2$ has

$$\Re\lambda_{\ell,+} = -\frac{\alpha}{n} - \frac{\ell^2k^2}{2n^2} + O(n^{-3}),$$

and the real part of its affected shift is again $-2(\beta-\alpha)n^{-2} + O(n^{-3})$, since the corresponding residue has real part 1/2. Hence its decay rate exceeds that of the first mode by $(\ell^2-1)k^2/(2n^2) + O(n^{-3})$. Modes with growing ℓ are farther left by the same dispersion bound, and the endpoint tumble mode has decay rate $2\alpha/n + O(n^{-2})$. This proves the required higher-mode ordering on both sides of $\alpha = k$.

At $\alpha = k$, the exact lattice splitting of the homogeneous first block is

$$\sqrt{\omega^2 - \sin^2(k/n)} = \frac{k^2}{\sqrt{3}\,n^2} + O(n^{-4}),$$

which proves (6.12) when the protected root wins. Near the center of this block, put

$$\lambda_c = \cos(k/n) - 1 - k/n, \qquad r_n = \sqrt{(k/n)^2 - \sin^2(k/n)}, \qquad \Delta = \lambda - \lambda_c.$$

The two $q = \pm k/n$ terms can be retained exactly, while the remaining terms are uniformly bounded for $|\Delta| \asymp n^{-3/2}$:

$$G_n(\lambda) = \frac{2(\Delta - k/n)}{n(\Delta^2 - r_n^2)} + O(1) = -\frac{2k}{n^2\Delta^2} + \frac{2}{n\Delta} + O(1). \tag{6.14}$$

For completeness, the bounded remainder follows by pairing wave numbers ℓ and $n-\ell$: for each fixed $\ell \neq 1$ its magnitude is bounded by a constant times $(\ell^2-1)^{-1}$, the sum of these bounds converges, and the terms with $|q_\ell|$ bounded away from zero form a bounded Riemann sum. For $\beta > k$, the equation $1 + 2\delta G_n = 0$ has the positive displacement

$$\Delta = 2\sqrt{k(\beta-k)}n^{-3/2} + O(n^{-2}),$$

which gives (6.13). For $\beta < k$, put $h = k - \beta > 0$ and substitute

$$\Delta = n^{-3/2}\big(y + xn^{-1/2}\big)$$

in $1 - 2hG_n/n = 0$. The constant and $n^{-1/2}$ coefficients in (6.14) give, respectively,

$$y^2 = -4kh, \qquad x = 2h.$$

Rouché's theorem on two disks of radius $Cn^{-5/2}$ about the resulting approximations, with fixed sufficiently large C, gives the affected pair

$$\Delta = \pm 2\mathrm{i}\sqrt{kh}\,n^{-3/2} + 2hn^{-2} + O(n^{-5/2}). \tag{6.15}$$

Its decay rate exceeds the protected decay rate by

$$\left(\frac{k^2}{\sqrt{3}} - 2h\right) n^{-2} + O(n^{-5/2}).$$

This coefficient is positive for $0 < h \le k$ because $k = 2\pi > 2\sqrt{3}$. The protected root therefore wins for every $\beta < k$; at $\beta = k$ the two parity roots coincide. This proves (6.12). At $\alpha = k$ and $\beta \le k$, the affected fixed mode $\ell \ge 2$ has decay rate

$$\frac{k}{n} + \left(\frac{\ell^2 k^2}{2} - 2(k-\beta)\right) n^{-2} O(n^{-3}).$$

Its order-n^{-2} coefficient exceeds the protected first coefficient by

$$\frac{(\ell^2-1)k^2}{2} + \frac{k^2}{\sqrt{3}} - 2(k-\beta) > 0.$$

For $\beta > k$, all these higher modes receive a negative real-part shift, while the positive $n^{-3/2}$ displacement of the first affected root dominates every n^{-2} correction. ■

7. Telegraph comparison

Definition 7.1 (Two continuum approximations)**.** For a homogeneous ring of circumference n, the standard telegraph system is

$$\partial_t \rho = -\partial_x m, \qquad \partial_t m = -\partial_x \rho - 2\omega m. \tag{7.1}$$

Its slow Fourier eigenvalue at wave number q is

$$\lambda^+_{\mathrm{tel}}(q) = -\omega + \sqrt{\omega^2 - q^2}. \tag{7.2}$$

The lattice-diffusion-corrected system is

$$\partial_t \rho = -\partial_x m + \tfrac{1}{2}\partial_x^2 \rho, \qquad \partial_t m = -\partial_x \rho + \tfrac{1}{2}\partial_x^2 m - 2\omega m. \tag{7.3}$$

whose slow eigenvalue is

$$\lambda^+_{\mathrm{td}}(q) = -\frac{q^2}{2} - \omega + \sqrt{\omega^2 - q^2}. \tag{7.4}$$

Theorem 7.2 *(Exact relaxation rate versus telegraph rates)***.** *Fix $\omega > 0$, let $q = 2\pi/n$, and define $\gamma_{\mathrm{tel}} = -\lambda^+_{\mathrm{tel}}(q)$ and $\gamma_{\mathrm{td}} = -\lambda^+_{\mathrm{td}}(q)$. Then*

$$\gamma_n^{\mathrm{hom}} - \gamma_{\mathrm{tel}} = \frac{q^2}{2} - \left(\frac{1}{24} + \frac{1}{6\omega}\right) q^4 + O(q^6), \tag{7.5}$$

$$\gamma_n^{\mathrm{hom}} - \gamma_{\mathrm{td}} = -\left(\frac{1}{24} + \frac{1}{6\omega}\right) q^4 + O(q^6). \tag{7.6}$$

Consequently

$$\frac{\gamma_{\mathrm{tel}}}{\gamma_n^{\mathrm{hom}}} \longrightarrow \frac{1}{1+\omega}, \qquad \frac{\gamma_{\mathrm{td}}}{\gamma_n^{\mathrm{hom}}} \longrightarrow 1. \tag{7.7}$$

In the bulk-critical scaling $\omega = \alpha/n$, the standard telegraph rate does recover the leading term:

$$\gamma_{\mathrm{tel}} = \frac{\alpha - \Re\sqrt{\alpha^2 - (2\pi)^2}}{n}, \tag{7.8}$$

which agrees with the leading terms of (6.10)–(6.13).

Proof. The exact lattice dispersion replaces $-q^2/2$ by $\cos q - 1$ and q inside the square root by $\sin q$. Expanding $\cos q$, $\sin^2 q$, and the square root gives (7.5)–(7.6). Dividing by the leading exact rate $(1+\omega)q^2/(2\omega)$ gives (7.7). Substitution of $\omega = \alpha/n$ and $q = 2\pi/n$ into (7.2) gives (7.8). ■

Proposition 7.3 *(Point-defect telegraph secular equation)***.** *On the circle cut at zero, define the point-defect telegraph operator by*

$$\mathcal{A}_\delta(\rho, m) = (-m', -\rho' - 2\omega m) \quad \text{on } (0, n),$$

with domain consisting of $m, \rho \in H^1(0,n)$ having traces that satisfy

$$m(0+) = m(n-), \qquad \rho(0+) - \rho(n-) = -2\delta m(0). \tag{7.9}$$

This is the distributional realization obtained by adding $-2\delta\,\delta_0(x)m$ to the second equation in (7.1)*. Outside the homogeneous telegraph spectrum, put*

$$\kappa = \sqrt{\lambda(\lambda + 2\omega)}$$

and interpret $\kappa^{-1}\coth(n\kappa/2)$ as the even meromorphic function of κ obtained by analytic continuation; no sign choice for the square root is required. The affected-parity eigenvalues satisfy

$$1 + \delta\frac{\lambda}{\kappa}\coth\left(\frac{n\kappa}{2}\right) = 0. \tag{7.10}$$

The reflection-even telegraph modes are unchanged. Formula (7.10) *is the continuum counterpart of* (3.8)*; the exact lattice formula retains $1 - \cos q$ and $\sin q$, and* (7.5)–(7.6) *quantify the resulting relaxation-rate error.*

Proof. The jump in (7.9) cancels the delta distribution in $-\rho' - 2\delta\delta_0 m(0)$, while continuity of m prevents a delta distribution in $-m'$. Thus the stated domain gives the point interaction on $L^2((0,n);\mathbb{C}^2)$. It is closed: the graph norm controls $\|m'\|_2$ directly and controls $\|\rho'\|_2$ by $\|\rho'\|_2 \le \|(\mathcal{A}_\delta(\rho,m))_2\|_2 + 2\omega\|m\|_2$; the trace theorem then preserves (7.9) under graph-norm limits.

The polarization entry of the telegraph resolvent is

$$\frac{1}{n}\sum_{\ell\in\mathbb{Z}} \frac{\lambda}{\lambda(\lambda+2\omega) + (2\pi\ell/n)^2} = \frac{\lambda}{2\kappa}\coth\left(\frac{n\kappa}{2}\right).$$

The series converges normally away from the homogeneous spectrum and the last expression supplies its meromorphic continuation when κ is purely imaginary. Applying the rank-one determinant identity gives (7.10). The defect vector is reflection odd, so the even modes are protected exactly as in Lemma 4.2. ■

8. Conclusions

Remark 8.1 (The theorem package). For the generator in Definition 2.1:

1. Theorem 3.3 and Theorem 3.4 give exact characteristic equations for every n, ω, and ω_0.
2. Theorems 4.7 and 4.8 classify the algebraic and geometric multiplicities of every eigenvalue, including every exceptional point.
3. Theorem 5.2 gives an exact criterion and decay length for a defect-localized mode.
4. Theorems 6.4 and 6.6 give the fixed-rate, vanishing-defect, and bulk-critical large-n gap asymptotics.
5. Theorem 7.2 and Proposition 7.3 compare the exact relaxation rate with the standard and diffusion-corrected telegraph approximations.

Each item retains the hypotheses stated in its cited theorem: in particular, localization is asserted off the bulk spectrum (with the compact case treated separately), finite-ring localization convergence assumes a simple noncompact root, and the gap results use their stated fixed-rate or double-scaling regimes.

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